

16 Ans: D

$$\begin{aligned}\text{Total heat supplied by the potatoes} &= \text{heat for calorimeter} + \text{heat for water} \\ &= C\theta + mc\theta \\ &= 200 \times 12 + (2000) \times (4.18) \times 12 \\ &= 102720 \text{ J}\end{aligned}$$

Hence heat supplied per gram of potato = $Q/m = 102720/75 = 1369.6 \text{ J per gram}$